



Mayan Kumar

B.Tech in Computer Science & Engineering
Cambridge Institute of Technology
Dhanbad, Jharkhand

+91-7667643249
kumarmayan92@gmail.com
github.com/Mayan-kr
linkedin.com/in/mayankumar01

EDUCATION

- **Cambridge Institute of Technology** 2022 – 2026
Bachelor of Technology in Computer Science & Engineering CGPA: 8.2/10.0

PERSONAL PROJECTS

- **RAG Arena (RAG Architecture Benchmarking)** Github link
Tools: Python, LlamaIndex, Streamlit, Groq, Neo4j Aura, ChromaDB, Ragas
 - Orchestrated dual LlamaIndex pipelines comparing an in-memory ChromaDB vector store against a cloud Neo4j Aura graph database.
 - Engineered an evaluation engine combining Ragas metrics with custom fallback scoring for rate-limit resilience.
 - Built a Streamlit dashboard with side-by-side generation, latency tracking, and a 70/30 quality-to-latency scoring algorithm.
- **Spider-ARM-Assistant (Agentic AI Automation)** Github link
Tools: Python, Qwen 1.5B, Firebase, WMI, GPUUtil
 - Developed an agentic system using fine-tuned Qwen 1.5B and a Python backend for real-time OS-level task automation.
 - Designed a secure Human-in-the-Loop (HITL) “Safety Handshake” via Firebase to validate sensitive system operations.
 - Achieved 98%+ tool accuracy on a 1.5B model via a custom regex-based JSON reconstruction engine to repair outputs.
 - Engineered a monitoring pipeline using WMI/GPUUtil to stream live system metrics to a remote mobile dashboard.
- **Amazon Sales Analysis** Github link
Performed EDA and built a recommendation system for sales data.
 - Cleaned and preprocessed raw sales data using Pandas, performing exploratory data analysis (EDA) to resolve formatting issues and prepare fields for accurate analysis.
 - Uncovered critical trends in pricing, discounts, and customer ratings to identify top-performing product categories and strategic sales insights.
 - Built a content-based product recommendation engine utilizing Scikit-Learn similarity metrics to suggest top-matching items to users.

EXPERIENCE

- **NIAMT** June 2025 – July 2025
AI/ML Intern Hatia (Hybrid)
 - Built content-based recommendation engine using TF-IDF and cosine similarity.
 - Performed data preprocessing and feature engineering to enhance recommendation accuracy.
 - Developed interactive Streamlit app for real-time movie search and suggestions.
- **Unified Mentor Pvt. Ltd.** May 2024 – June 2024
Data Analyst Intern Remote
 - Analyzed 100,000+ rows using Pandas to identify trends and improve marketing by 15%.
 - Performed EDA, built dashboards, and delivered insights for strategic decisions across 3 product categories.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C
AI & LLM Ops: PyTorch, Fine-tuning (QLoRA), RAG, Agentic Workflows, LangChain, LlamaIndex, Ragas, Quantization, HITL
Frameworks & Libraries: FastAPI, RESTful APIs, React, PySpark, Pandas, NumPy, Scikit-Learn, Matplotlib
Tools & Databases: Neo4j (Cypher), ChromaDB, Power BI, Docker, PostgreSQL, AWS, Oracle Cloud, Git, Databricks

ACHIEVEMENTS & CERTIFICATIONS

- **Neo4j Certified Professional** Neo4j Graph Database July 2025
- **Oracle Cloud Infrastructure 2025** Certified Foundations Associate Apr 2025
- **Intro to Machine Learning** Kaggle Certification May 2025
- **Oracle Dev Gym** Databases for Developers: Foundations Jan 2025

POSITIONS OF RESPONSIBILITY

- **President,** Python & Data Science Club 2025
- **Lead Anchor,** College Fests 2023-2025